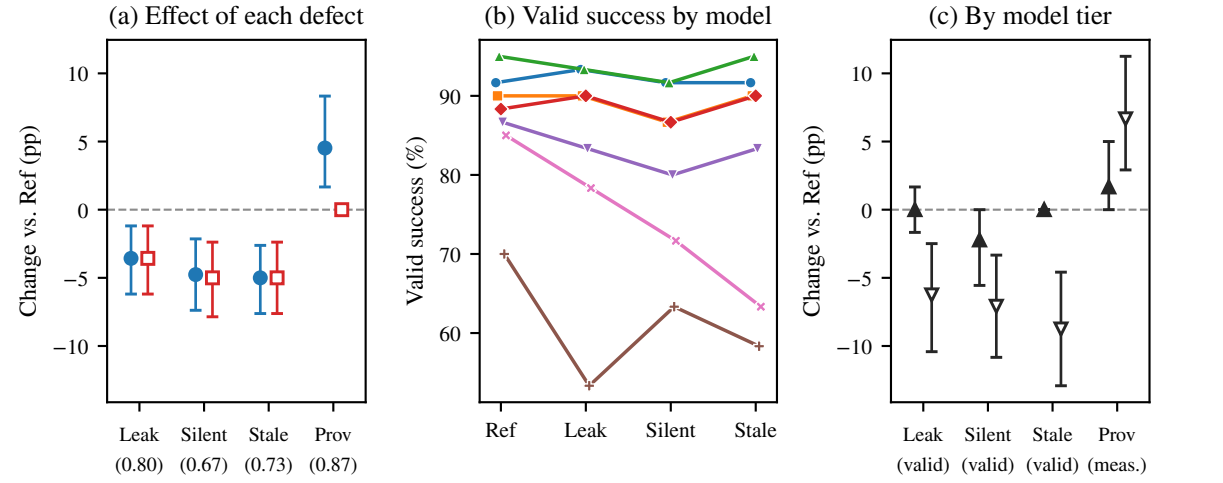
(a) Effect of each defect
Change vs. Ref (pp)
10
5
0
−5
−10
Leak
(0.80)
Silent
(0.67)
Stale
(0.73)
Prov
(0.87)
(b) Valid success by model
Valid success (%)
90
80
70
60
Ref
Leak
Silent
Stale
(c) By model tier
Change vs. Ref (pp)
10
5
0
−5
−10
Leak
(valid)
Silent
(valid)
Stale
(valid)
Prov
(meas.)
Measured
Valid
Frontier models
Efficient models
GPT-6.1-sol
Sonnet 5.5
Gemini 4 Argon
Haiku 5.5
GPT-6-luna
DeepSeek-V4.1-flash
Qwen3.8-flash